Searching

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- Finding an element, whether that is present in a list or not is called searching.

- It is the algorithmic process of finding a particular item in a collection of items.

- If a key is found in a list that is called successful type of search.

Searching Techniques

- **To search an element in a given array, it can be done in following ways:**

  1. Sequential Search (Linear Search)

  2. Binary Search

1. Sequential Search

- Sequential search is also called as Linear Search.
- Sequential search starts at the beginning of the list and checks every element of the list.
- It is a basic and simple search algorithm.
- Sequential search compares the element with all the other elements given in the list. If the element is matched, it returns the value index.

Example

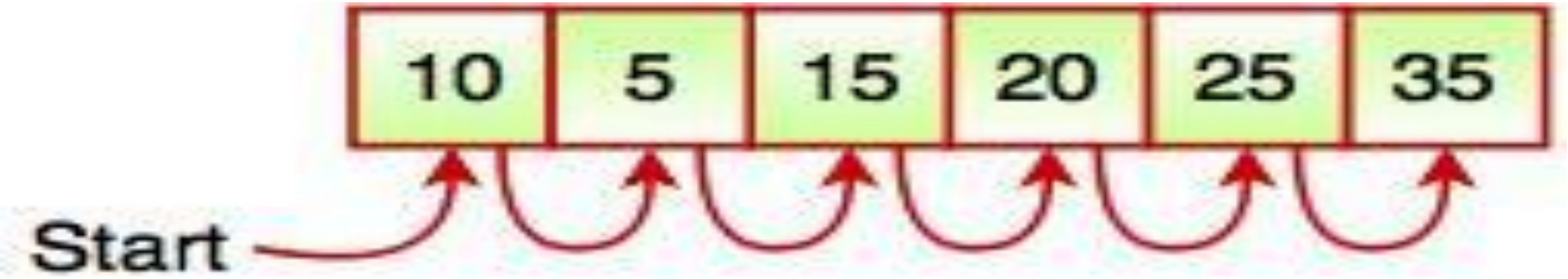

Fig. Sequential Search

Algorithm of Linear Search

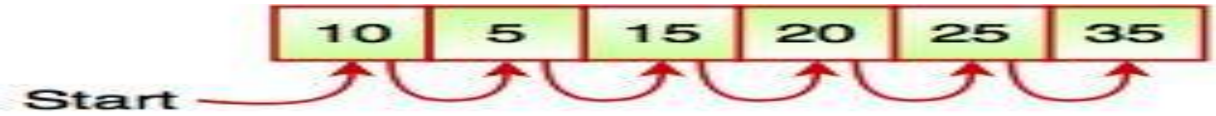

Fig. Sequential Search

linearSearch(a[maxsize],item)

Here a[maxsize] is an array of maxsize and item is an element which we wants to search in an array a . Here we take one variable loc =-1.

```
Step 1: Start

Step 2: for(i=0;i<=maxsize-1;i++)
            {

                 if (a[i] == item)     /* if required element found */
                    {
                           loc = i+1;
                           break;
                     }
               }
```

```
Step 3: if (loc !=-1)
            printf(" element is found at location %d", loc);
        else
            printf(" element is not found);

Step 4: Stop
```

Complexity of Linear Search

- In best case : O(1)

- In worst case: O(n), where n = total no of elements in an array.

- In average case: O((n+1)/2)

2. Binary Search

- Binary Search is used for searching an element in a sorted array.

- Binary search works on the principle of divide and conquer.

- This searching technique looks for a particular element by comparing the middle most element of the collection.

Cont…

- It is useful when there are large number of elements in an array.

Example:

5	10	15	20	25	30

- The above array is sorted in ascending order. As we know binary search is applied on sorted lists only for fast searching.

Cont…

- if searching an element 25 in above array , following figure shows how binary search works:

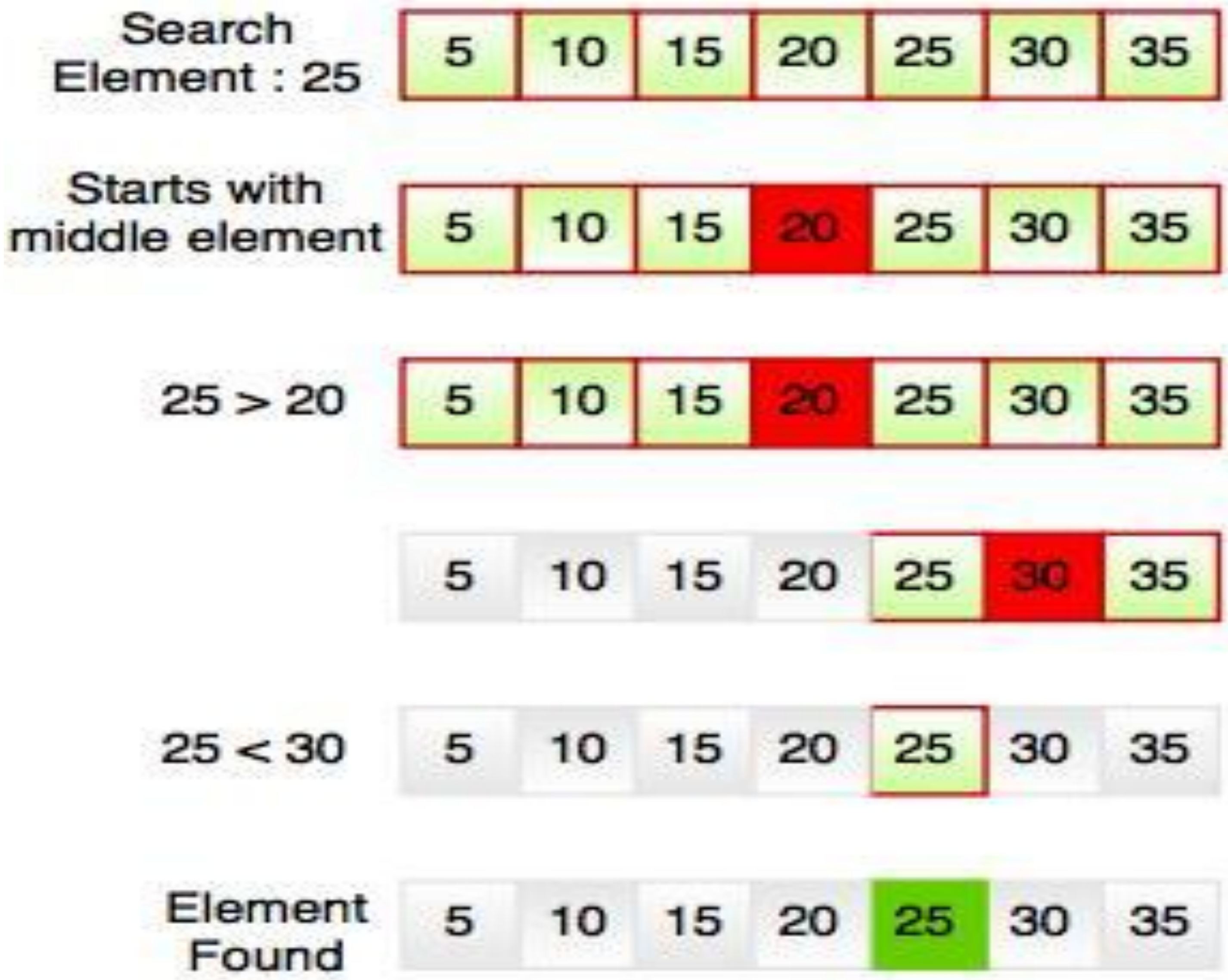

Fig. Working Structure of Binary Search

Binary_Search(a, lower_bound, upper_bound, val) // 'a' is the given array, 'lower_bound' is the index of the first array element, 'upper_bound' is the index of the last array element, 'val' is the value to search

Step 1: set beg = lower_bound, end = upper_bound, pos = - 1

Step 2: repeat steps 3 and 4 while beg **<**=end

Step 3: set mid = (beg + end)/2

Step 4: if a[mid] == val

set pos = mid

print pos

go to step 6

```
        else if a[mid] > val
                set end = mid - 1
        else
                set beg = mid + 1
        [end of if]
        [end of loop]
Step 5: if pos = -1
                print "value is not present in the array"
        [end of if]
Step 6: exit
```